Multimodal Learning for Understanding and Classifying Stress

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Introduction

StressID Dataset

- StressID[1] includes the following modalities: Video, audio, electrocardiography (ECG), electrodermal activity (EDA), and respiration signals.
- 65 participants are asked to complete 11 different tasks that should provoke varying levels of stress.
- The dataset relies on a self-assessment, which is done after each task, regarding the participants level of arousal, valence, stress and relax.

Research Question:

How does the different modalities interact in a multimodal setup and how can they be utilized in machine learning models to classify stress?

Challenges:

- Missing modalities.
- Unreliable subjects.

System Model

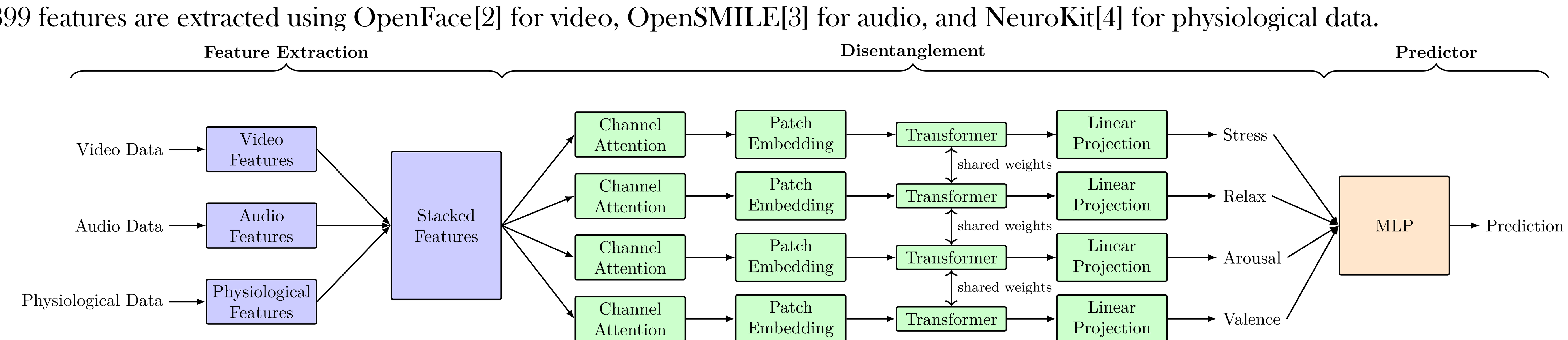
Disentangled Approaches

- Channel Attention masks $\mathcal{M}(x)$ are computed by: $\mathcal{M}(x) \coloneqq \sigma\big(W_1\, ReLU\big(W_0 BN\big(AvgPool(x)\big) + b_0\big) + b_1\big)$,

  where σ is the sigmoid activation function, $ReLU$ is the $ReLU$ activation function, BN is batch normalization, $AvgPool$ is global average pooling and W_1, W_2, b_0, b_1 are weights and biases.
- Backbone of the models are transformers with shared weights.

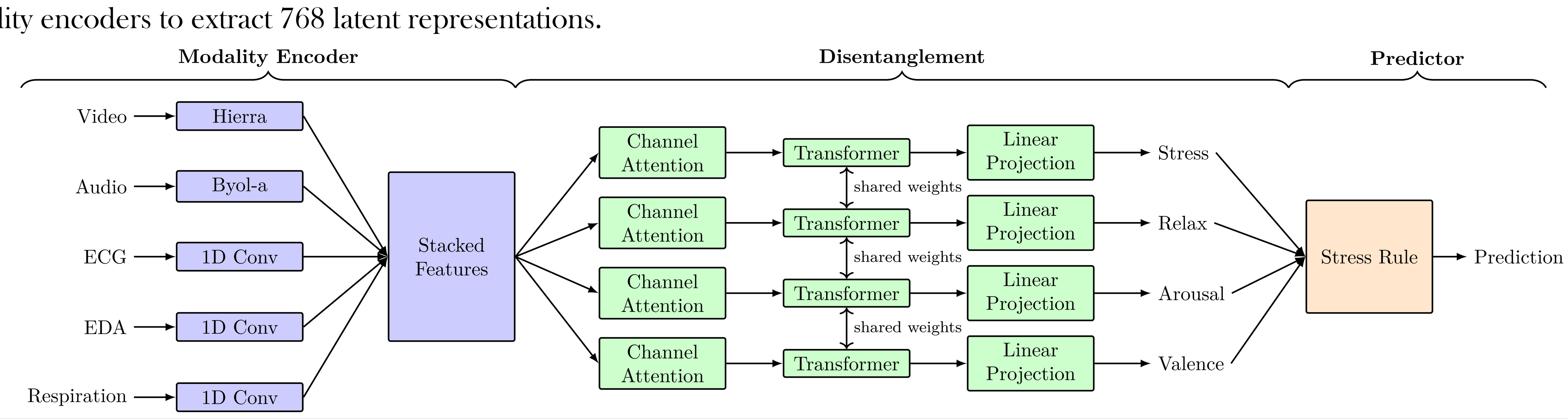
Summary Statistics (SumStat)

- A total of 399 features are extracted using OpenFace[2] for video, OpenSMILE[3] for audio, and NeuroKit[4] for physiological data.



Self-Supervised Model (SSM)

- Utilizes modality encoders to extract 768 latent representations.



Results

Performance

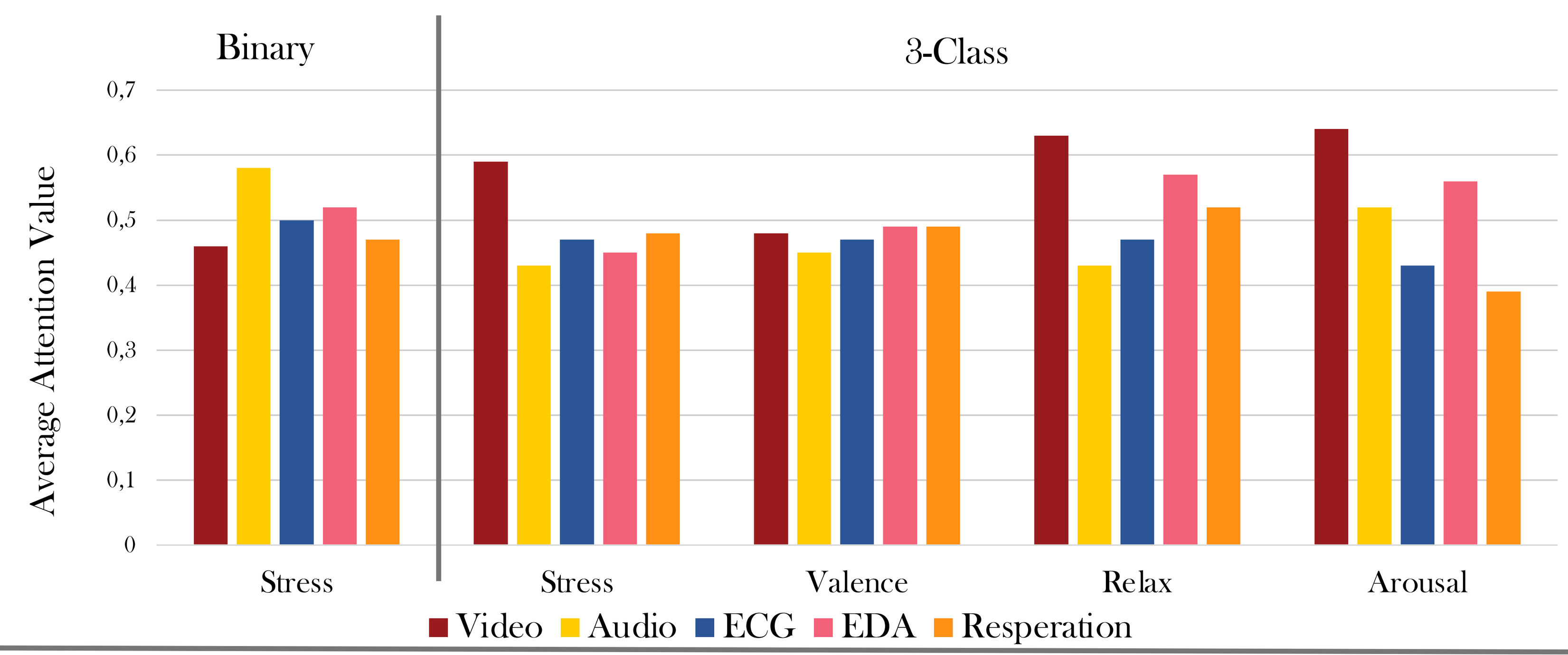
The performance of the different models are measured by the balanced accuracy and the weighted F1-score. The results are obtained from a 10-fold cross-validation.

	Binary Accuracy		Binary F1 Score		3-Class Accuracy		3-Class F1 Score	
	Seen	New	Seen	New	Seen	New	Seen	New
StressID	65(5)*	-	72(5)*	-	58(7)*	-	67(3)*	-
ADAPT	78.4(15)	57.6(16)	78.4(13.9)	58.7(16)	-	-	-	-
ShaSpec+	70.2(3.7)+	-	75.7(3.7)+	-	-	-	-	-
SumStat	59(4.1)	62(6.4)	59(4.1)	61.6(7)	40(4.9)	38.8(5.2)	65.5(5.1)	65.2(9.1)
Q-Former	66.4(3.2)	67.5(6.3)	65.9(3.5)	68(5.5)	43.5(7.2)	42.6(5.5)	65.7(4.1)	64.8(9)
FineTuned LLM	64.8(5.7)	62.2(7.4)	63.1(8.1)	62.5(8)	41.9(7.9)	37.3(6.3)	64.3(2.7)	65.9(10)
SSM	79.6(5.7)	82.4(8)	79.7(5.7)	82.9(7.7)	55.6(6.4)	67.4(9)	67(4)	73.5(12.7)

Table 1: The performance of the different models on seen and new subjects. (*) indicates that the performance is only for subjects where all modalities are present. (+) indicates the folds are split differently for that performance.

Interpretation

To better understand the performance of the models the value of the channel-attention are considered.



Statistical Significance

To determine if the models performs differently, the statistical significance is considered using Corrected Paired Student's t-test[5].

	ADAPT	Q-Former	FineTuned LLM	SumStat
Binary Seen	0.84	$1.04 \cdot 10^{-11}$	$2.30 \cdot 10^{-9}$	$9.07 \cdot 10^{-6}$
Binary New	0.01	$6.64 \cdot 10^{-9}$	$4.79 \cdot 10^{-8}$	$3.33 \cdot 10^{-3}$
3-class Seen	-	$6.38 \cdot 10^{-7}$	$1.95 \cdot 10^{-6}$	$9.40 \cdot 10^{-3}$
3-class New	-	$2.98 \cdot 10^{-6}$	$1.05 \cdot 10^{-6}$	$1.15 \cdot 10^{-4}$

Table 2: P-value for balanced accuracy comparing the self-supervised model with other models.

Future Work

Identifying Unreliable Subjects

- Leave-One-Out to identify if some subject cannot be predicted from the data.
- Incorporate a score of the reliability of a subject

Interaction Between Modalities

- Utilizing the SINC algorithm to determine what features are connected.
- Ablation study removing modalities.

Zero-shot Learning with LLM

- Investigate the performance of zero-shot learning with LLM.

References
[1] Chaptoukaev, H., et al. (2023). StressID: a Multimodal Dataset for Stress Identification.
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[3] Eyben, F., et al. (2010). "openSMILE - The Munich Versatile and Fast Open-Source Audio Feature Extractor", Proc. ACM Multimedia (MM), ACM, Florence, Italy, ISBN 978-1-60558-933-6, pp. 1459-1462
[4] Makowski, D., et al. (2021). "NeuroKit2: A Python toolbox for neurophysiological signal processing"
[5] Nadeau, C., & Bengio, Y. (1999)."Inference for the generalization error". Advances in neural information processing systems.

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